

Intermediate Microeconomics — Week 3, Day 2

In-class Exercises: Introduction to Game Theory

work in pairs • solutions at the end

Exercise 1 — Reading 2×2 games [competency: dominance, Nash equilibrium, game type]

For each game below the row player's payoff is listed first. Find all pure-strategy Nash equilibria, note any dominant strategies, and classify the game (Prisoner's Dilemma, coordination, or zero-sum).

Game A — two firms choosing Advertise (A) or Not (N).

	A	N
A	(5, 5)	(12, 3)
N	(3, 12)	(8, 8)

Game B — two firms choosing a common standard X or Y.

	X	Y
X	(3, 3)	(0, 0)
Y	(0, 0)	(2, 2)

Game C — a penalty: kicker (row) vs. goalie (column), payoff = kicker's success.

	Dive L	Dive R
Kick L	(0, 0)	(1, -1)
Kick R	(1, -1)	(0, 0)

- (a) Solve Game A. Is there a dominant strategy? Is the equilibrium efficient? Which classic game is this?
- (b) Solve Game B. How many pure equilibria are there? Which do the players prefer, and what is the coordination problem?
- (c) Solve Game C. Show there is no pure-strategy equilibrium.

Exercise 2 — Mixing and moving in sequence [competency: mixed NE and backward induction]

Part A — Mixed strategy. Use Game C above. Let the kicker play “Kick L” with probability p and the goalie play “Dive L” with probability q .

- (a) Choose q to make the kicker indifferent between Kick L and Kick R.
- (b) Choose p to make the goalie indifferent. State the mixed-strategy Nash equilibrium and the kicker's success probability in it.

Part B — Sequential game. A small firm (the **Entrant**) decides to enter (In) or stay Out. If it enters, the **Incumbent** chooses to Fight or Accommodate. Payoffs are (Entrant, Incumbent):

Out $\rightarrow$ (0, 10); In then Accommodate $\rightarrow$ (3, 4); In then Fight $\rightarrow$ (-2, 2).

- (c) Draw the game tree and solve it by backward induction.
- (d) Is “Fight” a credible threat? Explain what the incumbent would need in order to deter entry.

Reconvene: for each game state how many equilibria it has and of what type; for the tree, state the credible outcome.